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Impurity reduction in silicon-containing films

Abstract

Various embodiments herein relate to methods and apparatus for depositing doped and undoped silicon-containing films having a high degree of purity. In one example, the method includes exposing the substrate to a first reactant and a second reactant; reacting the first and second reactants with one another to form a silicon-containing material and depositing a portion of the silicon-containing film on the substrate; before the silicon-containing film is complete, performing an impurity reduction operation including: (i) generating a plasma from a plasma generation gas comprising inert gas and hydrogen, where the plasma generation gas is substantially free of oxygen, and (ii) exposing the substrate to the plasma to thereby reduce a concentration of fluorine, carbon, hydrogen, and/or nitrogen in the silicon-containing film; and repeating these operations (or a subset thereof) until the silicon-containing film is deposited to a final thickness.

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Background/Summary

INCORPORATION BY REFERENCE

(1) A PCT Request Form is filed concurrently with this specification as part of the present application. Each application that the present application claims benefit of or priority to as identified in the concurrently filed PCT Request Form is incorporated by reference herein in their entireties and for all purposes.

BACKGROUND

(2) Semiconductor device fabrication involves many different processes including, e.g., deposition and etching. One material that is commonly deposited is silicon oxide.

(3) The background description provided herein is for the purposes of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

(4) One aspect involves a method for depositing a doped or undoped silicon-containing film on a substrate, the method including: (a) exposing the substrate to a first reactant, where the first reactant is a silicon-containing reactant; (b) reacting at least the first reactant in a first plasma to form a doped or undoped silicon-containing material, and depositing a portion of the doped or undoped silicon-containing film on the substrate, the doped or undoped silicon-containing film having a first impurity concentration; (c) before the depositing of the portion of the doped or undoped silicon-containing film is complete, performing an impurity reduction operation including: (i) generating a second plasma from a plasma generation gas, where the plasma generation gas includes inert gas and hydrogen (H.sub.2), and where the plasma generation gas is substantially free of oxygen (O.sub.2), and (ii) exposing the substrate to the second plasma to thereby reduce the first impurity concentration to a second impurity concentration in the doped or undoped silicon-containing material; and (d) repeating (a) with at least one of (b) or (c) until the doped or undoped silicon-containing film is deposited to a final thickness.

(5) In various embodiments, the impurity is one or more of fluorine, carbon, hydrogen, nitrogen, and combinations thereof.

(6) In various embodiments, the second reactant is an oxygen-containing reactant.

(7) In some embodiments, the first reactant and second reactant are introduced to a chamber including the substrate simultaneously. In some embodiments, the first reactant and second reactant are introduced to a chamber including the substrate in temporally separated pulses.

(8) In various embodiments, the method also includes (e) exposing the substrate to a second reactant, such that reacting at least the first reactant includes reacting the first reactant with the second reactant to form the doped or undoped silicon-containing material, such that repeating (a) with at least one of (b) or (c) further includes repeating (e).

(9) In various embodiments, the doped or undoped silicon-containing film is deposited in recessed features formed in a surface of the substrate, the method further including exposing the substrate to etching chemistry to etch a top portion of the doped or undoped silicon-containing film in the recessed features before the doped or undoped silicon-containing film fully fills the recessed

features.

- (10) In various embodiments, the doped or undoped silicon-containing film is deposited in recessed features formed in a surface of the substrate, the method further including exposing the substrate to inhibition chemistry to selectively inhibit deposition near a top of the recessed features as compared to a bottom and middle of the recessed features.
- (11) In some embodiments, the second impurity concentration is less than 10 times the first impurity concentration.
- (12) In some embodiments, concentration of fluorine in the doped or undoped silicon-containing film is on the order of about $1E16$ atoms/cc once the doped or undoped silicon-containing film reaches the final thickness.
- (13) In some embodiments, concentration of carbon in the doped or undoped silicon-containing film is about $1E20$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness.
- (14) In some embodiments, concentration of carbon in the doped or undoped silicon-containing film is about $5E19$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness.
- (15) In some embodiments, concentration of carbon in the doped or undoped silicon-containing film is about 2% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness. For example, in some embodiments, the concentration of carbon in the doped or undoped silicon-containing film is about 0.5% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness.
- (16) In some embodiments, concentration of hydrogen in the doped or undoped silicon-containing film is about $5E20$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness. For example, in some embodiments, the concentration of hydrogen in the doped or undoped silicon-containing film is about $1.5E20$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness.
- (17) In some embodiments, concentration of hydrogen in the doped or undoped silicon-containing film is about 5% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness. For example, in some embodiments, the concentration of hydrogen in the doped or undoped silicon-containing film is about 0.75% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness.
- (18) In some embodiments, concentration of nitrogen in the doped or undoped silicon-containing film is about $3E20$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness.
- (19) In some embodiments, concentration of nitrogen in the doped or undoped silicon-containing film is about 1.5% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness.
- (20) In some embodiments, generating the plasma in (d)(i) includes flowing the inert gas at a rate of about 2 slm to about 60 slm, flowing the hydrogen at a rate of about 0.5 slm to about 5 slm, and generating the plasma at an RF power level that includes about 1000 W to about 5000 W HF RF, and about 0 W to about 2000 W LF RF.
- (21) In some embodiments, the substrate is exposed to the plasma in (d)(ii) for a duration of about 1 second or less.
- (22) In some embodiments, the inert gas in the plasma generation gas includes argon.
- (23) In some embodiments, the plasma generation gas includes nitrogen (N.sub.2).
- (24) In some embodiments, the first reactant includes a silane. In some embodiments, the first reactant includes aminosilane.
- (25) Another aspect involves an apparatus for depositing a doped or undoped silicon-containing film on a substrate, the apparatus including: a processing chamber; an inlet to the processing chamber for introducing reactants to the processing chamber; a plasma generator for generating a

plasma in the processing chamber; and a controller configured to cause any of the methods described herein.

(26) These and other aspects are described further below with reference to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates a process flow where silicon oxide film is deposited through atomic layer deposition, where impurity reduction is included as part of the atomic layer deposition cycle.
- (2) FIG. 1B illustrates a process flow where silicon oxide film is deposited through atomic layer deposition, where impurity reduction is periodically performed after a number of atomic layer deposition cycles.
- (3) FIG. 1C depicts a process flow where silicon oxide film is deposited through an inhibition-based deposition scheme, where impurity reduction is periodically performed over the course of depositing the silicon oxide film.
- (4) FIG. 1D illustrates a process flow where silicon oxide film is deposited through a deposition-etch-deposition scheme, where impurity reduction is periodically performed over the course of depositing the silicon oxide film.
- (5) FIG. 1E depicts a process flow where silicon oxide film is deposited through chemical vapor deposition, where impurity reduction is periodically performed over the course of depositing the silicon oxide film.
- (6) FIG. 2 shows an example process station that may be used to implement the methods described herein.
- (7) FIG. 3 illustrates an example multi-station processing tool that may be used to implement the methods described herein.
- (8) FIGS. 4A-4C depict experimental results related to the concentration of different impurities where the silicon oxide film was deposited using some methods.
- (9) FIGS. 5A and 5B depict experimental results related to the concentration of different impurities where the silicon oxide film was deposited without periodic impurity reduction (FIG. 5A) and with a periodic impurity reduction step (FIG. 5B).
- (10) FIG. 6 presents the concentration (in atomic percent) of various impurities, comparing a process without impurity reduction to a process where impurity reduction takes place.
- (11) FIGS. 7A and 7B show experimental FTIR data that suggest the impurity reduction described herein results in a decrease in hydrogen-related peaks (FIG. 7A) and an increase in silicon-oxygen related peaks (FIG. 7B).

DETAILED DESCRIPTION

(12) In the following description, numerous specific details are set forth to provide a thorough understanding of the presented embodiments. The disclosed embodiments may be practiced without some or all of these specific details. In other instances, well-known process operations have not been described in detail to not unnecessarily obscure the disclosed embodiments. While the disclosed embodiments will be described in conjunction with the specific embodiments, it will be understood that it is not intended to limit the disclosed embodiments.

(13) Silicon-containing materials are commonly deposited during fabrication of semiconductor devices. Example silicon-containing materials include silicon oxide, silicon oxynitride, silicon nitride, silicon carbonitride, silicon oxycarbide, silicon carbohydride. Silicon-containing materials may be doped or undoped. Doped materials include dopants such as boron and/or phosphorous. A number of different deposition processes are available. In many cases, vapor-based deposition processes are used. Example vapor-based deposition processes include, e.g., atomic layer deposition (ALD), plasma enhanced atomic layer deposition (PEALD), chemical vapor deposition

(CVD), plasma enhanced chemical vapor deposition (PECVD), remote plasma chemical vapor deposition (RPCVD), remote plasma atomic layer deposition (RPALD), inductively coupled plasma chemical vapor deposition (ICP-CVD) or high density plasma chemical vapor deposition (HDP-CVD), and others. In some cases, more complex deposition schemes may be used, which may include cycling deposition operations with other operations such as etching, inhibition, and/or passivation.

(14) Unfortunately, many of these deposition processes result in substantial incorporation of impurities into the silicon-containing film. These impurities can include, e.g., fluorine, carbon, hydrogen, and nitrogen. These elements may negatively affect various physical and electrical properties of the devices being formed. For instance, such impurities in silicon oxide can cause a relatively higher wet etch rate, which correlates with a lower quality silicon oxide film. Further, the presence of these impurities can negatively affect the leakage current and breakdown voltage in the devices, making such films unsuitable for electrical isolation purposes.

(15) The carbon, hydrogen, and nitrogen may be introduced into the silicon-containing film as a result of the precursors that are used for depositing the silicon-containing film. For instance, aminosilane-based precursors are commonly used in vapor deposition techniques. These aminosilane-based precursors include carbon, hydrogen, and nitrogen in addition to the desired silicon. Under typical deposition conditions for depositing silicon oxide, a substantial amount of this carbon, hydrogen, and nitrogen is undesirably incorporated into the silicon oxide film during deposition.

(16) Various operations can be taken to reduce the carbon, hydrogen, and nitrogen impurities. For example, the deposition conditions can be controlled to minimize incorporation of the impurities. In one particular example, the concentration of impurities in a silicon oxide film can be reduced by depositing at relatively high temperatures (e.g., about 900° C. or greater). However, high deposition temperatures can be overly limiting in some cases.

(17) In another example, a halogen-based silicon-containing precursor may be used in place of an aminosilane-based precursor. Example halogen-based silicon-containing precursors include, but are not limited to, silicon tetrachloride (SiCl_4), dichlorosilane (SiH_2Cl_2), diiodosilane (SiH_2I_2), etc. Many such halogen-based silicon-containing precursors contain little or no carbon, nitrogen, and/or hydrogen, particularly as compared to common aminosilane-based precursors such as tris(dimethylamino)silane (also referred to as 3DMAS, having the formula $\text{C}_6\text{H}_{19}\text{N}_3\text{Si}$). For example, although dichlorosilane and diiodosilane each include two hydrogens, the ratio of Si:H is relatively high at 1:2. By contrast, in tris(dimethylamino)silane, the ratio of Si:H is much lower at 1:19. Because the halogen-based silicon-containing precursors include relatively less carbon, nitrogen, and hydrogen than the aminosilane-based precursors, there is less incorporation of these elements into the silicon oxide film during deposition. However, halogen species are known to attack metals in the processing chamber where deposition takes place, and the metal that is etched from the processing chamber can be deposited on substrates being processed. As such, the halogen-based precursor methods are difficult to implement and can cause degradation of processing equipment and poor defect performance on substrates.

(18) The fluorine impurities in a silicon-containing film may originate from a number of different sources. In many cases, fluorine-containing chemistry is periodically used to clean the processing chamber in which deposition takes place. This cleaning may target removal of film buildup on interior chamber surfaces. Residual fluorine may be left on the chamber surfaces after a cleaning operation, and can end up in the silicon oxide film deposited in the chamber. Because the fluorine is introduced via the chamber cleaning operation, it is understood that this cleaning can lead to incorporation of fluorine in a silicon oxide film regardless of the deposition scheme used to form the silicon oxide film. Further, in these or other cases, a silicon-containing film may be deposited using a particular deposition scheme that involves exposing the substrate to NF_3 or other fluorine-containing chemistry. This fluorine-containing chemistry is especially useful in the context

of gapfill, where silicon oxide is deposited in recessed features that may have a high depth:width aspect ratio. These features can be very difficult to fill in a uniform manner without the formation of voids or seams within the features. In one example, a deposition-etch-deposition scheme (“dep-etch-dep”) is used, where deposition, etching, and/or passivation operations are cycled with one another as the feature is filled with a silicon-containing material. The etching ensures that the feature remains sufficiently open to allow further silicon-containing material deposition without closing off the features and forming voids. However, the etching operation(s) typically involve exposing the substrate to NF.sub.3 or other fluorine-containing chemistry, which results in the incorporation of fluorine into the silicon-containing film.

(19) In another example, an inhibition-based deposition scheme is used, where inhibition, deposition, and/or passivation operations are cycled with one another as the feature is filled with a silicon-containing material. This deposition scheme may be referred to as inhibitor-controlled exposure (ICE) deposition. The inhibition involves exposing the substrate to NF.sub.3 or similar chemistry to selectively inhibit deposition at or near the top of the feature, thus promoting a bottom-up fill mechanism that fills the feature without the formation of voids or seams. While both the deposition-etch-deposition scheme and the inhibition-based deposition scheme involve periodic exposure to NF.sub.3 or similar chemistry, it is understood that the NF.sub.3 may have different effects (e.g., etching vs. inhibition) depending on the other processing conditions that are used.

(20) In either case, exposure of the substrate to NF.sub.3 or other fluorine-containing chemistry results in the incorporation of fluorine impurities in the silicon-containing film. In order to reduce the fluorine impurities, a passivation operation without impurity reduction may be used, as mentioned above, which involves exposing the substrate to plasma generated from argon, hydrogen (H.sub.2), and oxygen (O.sub.2). The plasma typically has a relatively high concentration of oxygen. In one example, the flow rate for a 4-station chamber of H.sub.2 is about 2 slm, the flow rate for a 4-station chamber of O.sub.2 is about 2 slm, and the flow rate for a 4-station chamber of Ar is about 10 slm. The plasma is typically generated at an RF power about 1250 W to about 2500 W (e.g., for a chamber having four stations each for processing a 300 mm substrate), and a substrate is exposed to the plasma for a duration about 5 to about 10 seconds during each iteration. The pressure in a reaction chamber for a 4-station chamber may be about 1 Torr to about 10 Torr, or about 2 Torr. When this passivation operation without impurity reduction is used during deposition, the fluorine content in the film is lower than it would be without the passivation operation. For instance, in various cases where the silicon-containing material is deposited through the deposition-etch-deposition mechanism or the inhibition-based deposition mechanism, the passivation operation may lower the fluorine content in the film by about one or two orders of magnitude. As an example, typical fluorine concentrations for such films are about 1E18 to about 1E21 atoms/cc, or about 1E18 to about 1E20 atoms/cc, with the highest fluorine concentrations often being found at interfaces/film depths where the deposition operations were stopped and etching or inhibition operations were used. Experimental results are described further below.

(21) The reduction in fluorine content achieved with the passivation operation without impurity reduction is beneficial. However, even with passivation, the fluorine concentration is higher than desired for many applications, and it would be advantageous to further reduce the fluorine content of the silicon-containing film.

(22) Provided herein are methods and apparatuses for incorporating an impurity reduction operation, sometimes referred to as a passivation operation, which can be used to substantially reduce the fluorine content in the film. Certain disclosed embodiments use processing conditions different from that of a passivation operation without impurity reduction such as described above. For instance, an impurity reduction operation as described herein can further reduce the fluorine content in the film by an additional one to two orders of magnitude compared to the passivation operation without impurity reduction. In various cases, deposition using the impurity reduction operation may produce a silicon oxide film having a fluorine content on the order of about 1E16

atoms/cc.

(23) The impurity reduction operation may be practiced in the context of depositing a silicon-containing material in recessed features (e.g., low aspect ratio features such as those having a depth:width aspect ratio as low as about 1:1, high aspect ratio features such as those having a depth:width ratio as high as about 100:1, and recessed features having aspect ratios of about the range of these values). The impurity reduction operation may also be practiced in the context of depositing a blanket film of a silicon-containing material, such as silicon oxide. Example applications include SiO₂ (in some cases SiO₂/SiN) deposition in NAND, DRAM, logic, PCRAM, and MRAM applications, etc.

(24) Advantageously, the impurity reduction operation also has the effect of reducing the carbon content, hydrogen content, and nitrogen content in silicon-containing films. As such, the impurity reduction operation produces silicon-containing films that are substantially purer than was previously achievable using aminosilane-based precursors and/or the deposition techniques described above.

(25) Further, the impurity reduction operation can be performed in substantially less time than is required for the passivation operation without impurity reduction. In other words, the impurity reduction operation achieves better results (e.g., greater reduction in impurities) in less time. As a result, individual cycle times and overall deposition times are significantly reduced, leading to increased throughput on certain deposition apparatuses. This is a substantial improvement.

Deposition Methods

(26) Certain disclosed embodiments may involve ALD. ALD is a technique that deposits thin layers of material using sequential self-limiting reactions. ALD processes use surface-mediated deposition reactions to deposit films on a layer-by-layer basis in cycles. As an example, an ALD cycle may include the following operations: (i) delivery/adsorption of a precursor, (ii) purging of precursor from the chamber, (iii) delivery of a second reactant and optionally ignite plasma, and (iv) purging of byproducts from the chamber. The reaction between the second reactant and the adsorbed precursor to form a film on the surface of a substrate affects the film composition and properties, such as nonuniformity, stress, wet etch rate, dry etch rate, electrical properties (e.g., breakdown voltage and leakage current), etc.

(27) Unlike a chemical vapor deposition (CVD) technique, ALD processes use surface-mediated deposition reactions to deposit films on a layer-by-layer basis. In one example of an ALD process, a substrate surface that includes a population of surface active sites is exposed to a gas phase distribution of a first precursor, such as a silicon-containing precursor, in a dose provided to a chamber housing a substrate. Molecules of this first precursor are adsorbed onto the substrate surface, including chemisorbed species and/or physisorbed molecules of the first precursor. It should be understood that when a compound is adsorbed onto the substrate surface as described herein, the adsorbed layer may include the compound as well as derivatives of the compound. For example, an adsorbed layer of a silicon-containing precursor may include the silicon-containing precursor as well as derivatives of the silicon-containing precursor. After a first precursor dose, the chamber is then evacuated to remove most or all of first precursor remaining in gas phase so that mostly or only the adsorbed species remain. In some implementations, the chamber may not be fully evacuated. For example, the reactor may be evacuated such that the partial pressure of the first precursor in gas phase is sufficiently low to mitigate a reaction. A second reactant, such as an oxygen-containing gas, is introduced to the chamber so that some of these molecules react with the first precursor adsorbed on the surface. In some processes, the second precursor reacts immediately with the adsorbed first precursor. In other embodiments, the second reactant reacts only after a source of activation is applied temporally. The chamber may then be evacuated again to remove unbound second reactant molecules. As described above, in some embodiments the chamber may not be completely evacuated. Additional ALD cycles may be used to build film thickness.

(28) In some implementations, the ALD methods include plasma activation. As described herein,

the ALD methods and apparatuses described herein may be conformal film deposition (CFD) methods, which are described generally in U.S. patent application Ser. No. 13/084,399 (now U.S. Pat. No. 8,728,956), filed Apr. 11, 2011, and titled "PLASMA ACTIVATED CONFORMAL FILM DEPOSITION," and in U.S. patent application Ser. No. 13/084,305, filed Apr. 11, 2011, and titled "SILICON NITRIDE FILMS AND METHODS," which are herein incorporated by reference in their entireties.

(29) FIGS. 1A-1E illustrate a number of deposition schemes that may be used in various embodiments. Each example deposition scheme can be performed alone or in combination with any one or more of the other deposition schemes depicted in FIGS. 1A-1E. FIG. 1A depicts an ALD cycle **101A** that is modified to include the impurity reduction operation. The x-axis represents time. In this example, the ALD cycle **101A** includes a dose operation **102**, followed by an optional purge operation **103**, followed by an oxidation operation **104**, followed by an impurity reduction operation **105**, followed by a second optional purge operation **106**. In ALD without impurity reduction, the impurity reduction operation **105** is omitted.

(30) When practicing the embodiment of FIG. 1A, a first reactant (e.g., a silicon-containing precursor, such as an aminosilane-based precursor) is provided to the reaction chamber during the dose operation **102**. The first reactant is allowed to adsorb onto a surface of the substrate. Excess first reactant may be removed from the reaction chamber during the purge operation **103**. This can be done by purging and/or evacuating the process chamber to remove excess first reactant, without removing the adsorbed layer. Purging can be done using any inert gas such as nitrogen (N.sub.2), argon (Ar), neon (Ne), helium (He), hydrogen (H.sub.2), and their combinations.

(31) A second reactant (e.g., an oxygen-containing reactant) is provided to the substrate during the oxidation operation **104**, and the first and second reactants react with one another on the surface of the substrate to form a silicon-containing material. In some cases, thermal energy is used to drive the reaction between the first and second reactants. In other cases, plasma energy may be used to drive this reaction.

(32) The substrate is exposed to an impurity reduction plasma during the impurity reduction operation **105**, as discussed further below. Excess reactants and byproducts are removed from the reaction chamber during the second purge operation **106**. Second purge operation **106** may involve any of the process conditions described above with respect to purge operation **103**. The purge operation **103** and second purge operation **106** may involve evacuating and/or sweeping the reaction chamber with inert gas. The ALD cycle **101A** of FIG. 1A may be repeated any number of times (e.g., X times in FIG. 1A) to achieve a desired film thickness.

(33) In a similar embodiment, the impurity reduction operation **105** may occur at a different time during the ALD cycle **101A** of FIG. 1A. In general, the impurity reduction operation **105** may occur before, during, or after the purge operation **106**, before a subsequent iteration of the dose operation **102**.

(34) FIG. 1B illustrates another example where deposition occurs through ALD. In this example, the ALD cycle **101B** includes the dose operation **102**, purge operation **103**, oxidation operation **104**, and second purge operation **106**. These operations are analogous to those described above, and they may be repeated any number of times (e.g., X times in FIG. 1B) to build up additional film thickness. After the ALD cycle **101B** is repeated X times, the impurity reduction operation **105** is performed. The ALD cycle **101B** and the impurity reduction operation **105** (see reference numeral **150**) may together be repeated any number of times (e.g., Y times in FIG. 1B). Performing the impurity reduction operation **105** more frequently may result in greater impurity reduction, but it also results in increased processing times and decreased throughput. In various embodiments where deposition occurs according to FIG. 1B, X may be about 1 to about 20, and Y may be about 1 to about 100. Y may be selected to achieve a desired final film thickness. In a particular embodiment of the method in FIG. 1B, the deposition rate is on the order of about 1 Å/cycle, X=20, and Y=100, producing a film that has a total thickness of approximately 2000 Å.

(35) In these or other embodiments, the impurity reduction operation **105** may occur periodically, with the frequency based on the thickness of film deposited in each cycle. In one example, the impurity reduction operation **105** may be repeated after each 1 nm of silicon-containing film is deposited (e.g., deposit 1 nm of silicon-containing film, perform impurity reduction **105**, repeat). More generally, the impurity reduction operation **105** may be repeated after each 0.1-2 nm portion/thickness of silicon-containing film is deposited. Where the silicon-containing film is deposited as a blanket film, the thickness of the blanket film may be used. Where the silicon-containing film is deposited in a recessed feature, the thickness of film may be measured from the bottom of the feature upwards.

(36) FIG. 1C depicts an example where deposition occurs through an inhibition-based mechanism (e.g., ICE), for example in the context of gapfill. The process starts with inhibition operation **110**, where the substrate is exposed to NF.sub.3 or similar chemistry to selectively inhibit deposition near the top of recessed features on the substrate surface, as compared to the bottom and middle of the recessed features. Next, ALD cycle **101B** is performed, including dose operation **102**, purge operation **103**, oxidation operation **104**, and purge operation **106**. ALD cycle **101B** may be repeated any number of times (e.g., X times in FIG. 1C). Next, the impurity reduction operation **105** is performed, as described further below. The impurity reduction operation **105** may be performed in place of a passivation operation without impurity reduction. In the example of FIG. 1C, the impurity reduction operation **105** also acts to completely remove any remaining inhibitor species present on the substrate, which originated from the inhibition operation **110**. The inhibitor species are slowly eroded away during the ALD cycles **101B**; however, some inhibitor species typically remain. By completely stripping away any remaining inhibitor species, the impurity reduction operation **105** restores the sidewalls of the feature (e.g., particularly the upper sidewalls where the inhibition species were present) back to their original uninhibited state. This ensures that a subsequent inhibition operation **110** can be performed in a uniform, reproducible manner, targeting the upper sidewalls for inhibition.

(37) The inhibition operation **110**, ALD cycle **101B**, and impurity reduction operation **105** (see reference numeral **152**) may be repeated together any number of times (e.g., Z times in FIG. 1C). As with the mechanism of FIG. 1B, increasing the frequency of the impurity reduction operation **105** may result in greater impurity reduction at the expense of longer processing times and lower throughput. In various embodiments where the deposition occurs according to the method shown in FIG. 1C, X may be about 1 to about 10, and Z may be about 1 to about 100. In these or other embodiments, the impurity reduction operation **105** may occur periodically, with the frequency based on the thickness of film deposited in each cycle. In one example, the impurity reduction operation **105** may be repeated after each 1 nm of silicon-containing film is deposited (e.g., deposit 1 nm of silicon-containing film, perform impurity reduction **105**, repeat). More generally, the impurity reduction operation **105** may be repeated after each about 0.1 nm to about 20 nm or about 0.1 nm to about 10 nm portion/thickness of silicon-containing film is deposited. Where the silicon-containing material is deposited in recessed features, the thickness of the silicon-containing film may be measured from the bottom of the feature upwards.

(38) FIG. 1D illustrates an example where deposition occurs through a deposition-etch-deposition mechanism, for example in the context of gapfill. The process starts with ALD cycle **101B**, which may be performed X times. Next, an etch operation **120** is performed to partially etch the silicon-containing material previously deposited in the recessed features to thereby ensure that the features do not become closed/clogged over the course of the deposition. The etch operation **120** may involve exposing the substrate to NF.sub.3 or other etching chemistry (e.g., often fluorine-containing chemistry), which may be provided in plasma form in various embodiments. The ALD cycle **101B** and the etch operation **120** (see reference numeral **153**) may together be repeated any number of times (e.g., Y times in FIG. 1D). Next, an impurity reduction operation **105** is performed, as described further below. The ALD cycle **101B**, etch operation **120**, and impurity

reduction operation **105** (see reference numeral **154**) may together be repeated any number of times (e.g., Z times in FIG. **1D**). In various embodiments where deposition occurs according to the method shown in FIG. **1D**, X may be about 1 to about 1000, Y may be about 1 to about 1000, and Z may be about 2 to about 5. X may be the same or different from Y. In many cases, X is greater than Y such that a majority (e.g., at least about 50%, at least about 75%, or at least about 90%) of the silicon-containing film is deposited before the etch operation **120** and the impurity reduction operation **105** take place. In various cases, X may be determined based on the size of the feature being filled.

(39) In these or other embodiments, the impurity reduction operation **105** may occur periodically, with the frequency based on the thickness of film deposited in each cycle. Generally, the impurity reduction operation **105** may be repeated after each about 0.1 nm to about 20 nm or about 0.1 nm to about 10 nm portion/thickness of silicon-containing film is deposited. Where the silicon-containing film is deposited in a recessed feature, the thickness of the film may be measured from the bottom of the feature upwards.

(40) FIG. **1E** illustrates an example where deposition occurs through chemical vapor deposition. The method begins with reactant delivery operation **130**, where a silicon-containing reactant and an oxygen-containing reactant are simultaneously delivered to the reaction chamber, where they react with one another and deposit silicon-containing material on the surface of the substrate. The reaction may be driven by thermal energy or plasma energy. Next, the impurity reduction operation **105** is performed, as discussed further below. The reactant delivery operation **130** and the impurity reduction operation **105** (see reference numeral **155**) may be repeated any number of times (e.g., X times in FIG. **1E**). In various embodiments where the deposition occurs according to the method shown in FIG. **1E**, X may be about 1 to about 1000, for example about 1 to about 100 or about 1 to about 10. In these or other embodiments, the impurity reduction operation **105** may occur periodically, with the frequency based on the thickness of film deposited in each cycle. In one example, the impurity reduction operation **105** may be repeated after each 10 nm of silicon-containing film is deposited (e.g., deposit 10 nm of silicon-containing film, perform impurity reduction **105**, repeat). More generally, the impurity reduction operation **105** may be repeated after each about 0.1 nm to about 20 nm or about 0.1 nm to about 10 nm portion/thickness of silicon-containing film is deposited.

(41) While ALD and CVD embodiments are described above, it will be understood that certain disclosed embodiments may be implemented with any deposition process that involves plasma. Non-limiting examples include RPCVD, RPALD, ICP-CVD, and HDP-CVD. In various embodiments, only a single reactant may be used to form a film in a plasma-based environment and an impurity reduction operation can still be used to reduce impurities in the film.

(42) Process Conditions for Impurity Reduction Operation

(43) In various embodiments, the impurity reduction operation involves exposing the substrate to plasma generated from a plasma generation gas that includes inert gas and hydrogen (e.g., H.sub.2). The plasma generation gas is substantially free of oxygen (e.g., O.sub.2 and other oxygen-containing species). As used herein, a plasma generation gas that is “substantially free” of oxygen may have trace amounts of oxygen; however, oxygen and oxygen-containing species are not intentionally provided as part of the plasma generation gas. In various embodiments, the inert gas includes argon. Alternatively or in addition, other noble gases (e.g., helium, neon, krypton, etc.) may be used in some cases. In some cases, the plasma generation gas may further include nitrogen (e.g., N.sub.2). Example flow rates for a 4-station chamber for the inert gas may be about 2 to about 60 slm. In some cases, the flow rate for the inert gas may be at least about 2 slm, at least about 5 slm, at least about 10 slm, at least about 20 slm, at least about 30 slm, or at least about 40 slm. In these or other cases, the flow rate of inert gas may be about 60 slm or less, about 50 slm or less, about 40 slm or less, about 30 slm or less, or about 20 slm or less. Example flow rates for the hydrogen (e.g., H.sub.2) may be about 0.5 and about 5 slm. In some cases, the flow rate of

hydrogen may be at least about 0.5 slm, at least about 0.1 slm, at least about 0.5 slm, at least about 1 slm, at least about 2 slm, or at least about 3 slm. In these or other cases, the flow rate of hydrogen may be about 5 slm or less, about 4.5 slm or less, about 4 slm or less, or about 3.5 slm or less. Where it is used, example flow rates for nitrogen (e.g., N.sub.2) may be up to about 30 slm, in some cases up to about 20 slm, or up to about 10 slm.

(44) The plasma may be generated at an RF power that includes about 1000 and about 6000 or about 1000 and about 5000 W (HF RF for a 13.56 MHz capacitively coupled plasma), and about 0 W and about 5000 W or 0 W and about 3000 W or about 0 W and about 2000 W (LF RF for a 400 kHz (or 50 kHz to 2 MHz) plasma) for a chamber having 4 stations. In other words, the RF power may be provided at single frequency (HF only) or dual frequency (e.g., HF and LF). In some cases, the HF RF for a 13.56 MHz capacitively coupled plasma may be provided at a power level of at least about 1000 W, or at least about 2000 W, or at least about 3000 W for a chamber having 4 stations. In these or other cases, the HF RF for a 13.56 MHz capacitively coupled plasma may be provided at a power level of about 6000 W or less, about 5000 W or less, in some cases about 4000 W or less, or about 3000 W or less for a chamber having 4 stations. In these or other cases, the LF RF may be omitted. In other embodiments, the LF RF for a 400 kHz (or 50 kHz to 2 MHz) plasma may be provided at a power level of at least about 1 W, at least about 10 W, at least about 50 W, at least about 100 W, at least about 500 W, or at least about 1000 W for a chamber having 4 stations. In these or other cases, the LF RF for a 400 kHz (or 50 kHz to 2 MHz) plasma may be provided at a power level of about 5000 W or less, or about 3000 W or less, or about 2000 W or less, for example 1500 W or less, or 1000 W or less, or 500 W or less, or 200 W or less for a chamber having 4 stations. These power levels are appropriate for a substrate having a diameter of about 300 mm, and may be scaled accordingly for substrates of different sizes. Example frequencies include, e.g., 13.56 MHz, 27 MHz, and 60 MHz, though these are not intended to be limiting. During the plasma exposure operations, the plasma may have a duty cycle about 10% and about 100%. The plasma may be any type of plasma. In some cases, the plasma is a capacitively coupled plasma. In other cases, the plasma is an inductively coupled plasma or a microwave plasma. In some embodiments, the plasma is a remote plasma. In some embodiments, the plasma is generated by electron cyclotron resonance (ECR).

(45) During the impurity reduction operation, the pressure in the reaction chamber may be maintained about 0.5 to about 20 Torr. In various embodiments, the pressure may be at least about 0.5 Torr, at least about 1 Torr, at least about 5 Torr, or at least about 10 Torr. In these or other cases, the pressure may be about 20 Torr or less, or about 15 Torr or less, or about 10 Torr or less. The substrate may be positioned on a substrate support, which may be temperature controlled. In some cases, the substrate may be heated or cooled during the impurity reduction operation, for example through the substrate support. Example substrate support temperatures may be about -400° C. to about 1400° C. or about -40° C. to about 1000° C. or about 150° C. to about 650° C. In some cases, the substrate support is maintained at a temperature of at least about 150° C., or at least about 200° C., or at least about 300° C., or at least about 400° C. In these or other cases, the substrate support may be maintained at a temperature of about 650° C. or less, or about 600° C. or less, or about 500° C. or less, or about 400° C. or less, or about 300° C. or less, or about 200° C. or less.

(46) During the impurity reduction operation, the substrate may be exposed to the plasma for a duration about 0.2 and about 120 seconds. In various embodiments, the plasma exposure duration may be at least about 0.2 seconds, or at least about 0.5 seconds, or at least about 0.1 second, or at least about 0.2 seconds. In these or other embodiments, the plasma exposure duration may be about 120 seconds or less, or about 10 seconds or less, or about 1 second or less, or about 0.5 seconds or less. In many cases, the desired impurity reduction is achieved very quickly, typically in less than 1 second. This is a substantial improvement (e.g., reduced processing time) over the existing passivation methods described above, which require at least 5-10 seconds of exposure time.

(47) In many cases, the impurity reduction operation may be performed at significantly higher flow rates for hydrogen and inert gas, as compared to a passivation operation without impurity reduction. Another difference between the impurity reduction operation and the passivation operation without impurity reduction is the omission of oxygen in the plasma generation gas. In the passivation operation without impurity reduction, oxygen is provided at a substantial flow rate. By contrast, in the impurity reduction operation, oxygen is omitted from the plasma generation gas. Further, the impurity reduction operation typically occurs at higher RF power levels than passivation operation without impurity reduction. As a result of such differences, the impurity reduction operation is substantially more effective in removing fluorine from the silicon-containing film (e.g., about one to two orders of magnitude more effective, in less time), as compared to the passivation operation without impurity reduction, as discussed further below in the Experimental section. Advantageously, the impurity reduction operation also lowers the concentration of carbon, hydrogen, and nitrogen in the film.

(48) Another advantage of the disclosed impurity reduction operation is that it can be performed in situ. In other words, it can be performed in the same reaction chamber in which the silicon-containing film is being deposited. While it is possible for the impurity reduction operation to take place in another chamber separate from the deposition chamber (e.g., with the substrate being transferred between chambers as needed), there is no need to provide a separate chamber for this purpose. Appropriate apparatus are discussed further below in the Apparatus section.

(49) Process Conditions for Deposition Operations

(50) As explained in relation to FIGS. 1A-1E, the impurity reduction operation described herein may be practiced in many different contexts using various reaction mechanisms. As such, the embodiments herein are not intended to be limited to a particular set of deposition conditions. However, certain example conditions are provided as guidance.

(51) In various embodiments where the deposition occurs through a method that involves atomic layer deposition cycles (e.g., as described in relation to FIGS. 1A-1D), one or more of the following reaction conditions may be used. During the dose operation, a silicon-containing reactant may be provided to the reaction chamber at a flow rate of about 150 sccm to about 900 sccm. The silicon-containing reactant is allowed to adsorb onto the surface of the substrate. The silicon-containing reactant may be a silane, and in many cases is an aminosilane. Example aminosilanes include, but are not limited to, tris(dimethylamino)silane, bis(tert-butylamino)silane, 1,1,1,3,3,3-hexamethyldisilazane, cyclic azasilanes, tetrakis(dimethylamino)silane, trisilylamine, bis(diethylamino)silane, etc. An aminosilane includes at least one nitrogen atom bonded to a silicon atom, but may also contain hydrogens, oxygens, halogens and carbons. Examples of aminosilanes are mono-, di-, tri- and tetra-aminosilane ($\text{H.sub.3Si(NH.sub.2).sub.4}$, $\text{H.sub.2Si(NH.sub.2).sub.2}$, $\text{HSi(NH.sub.2).sub.3}$ and $\text{Si(NH.sub.2).sub.4}$, respectively), as well as substituted mono-, di-, tri- and tetra-aminosilanes, for example, t-butylaminosilane, methylaminosilane, tert-butylsilanamine, bis(tertiarybutylamino)silane ($\text{SiH.sub.2(NHC(CH.sub.3).sub.3).sub.2}$ (BTBAS)), tert-butyl silylcarbamate, $\text{SiH(CH.sub.3)—(N(CH.sub.3).sub.2).sub.2}$, $\text{SiHCl—(N(CH.sub.3).sub.2).sub.2}$, $\text{(Si(CH.sub.3).sub.2NH).sub.3}$, di(sec-butylamino)silane (DSBAS), di(isopropylamido)silane (DIPAS), bis(diethylamino)silane (BDEAS), and the like. A further example of an aminosilane is trisilylamine ($\text{N(SiH.sub.3).sub.3}$). Various other silanes may be used, as well.

(52) Example silicon-containing reactants include, but are not limited to, silanes, polysilanes, halosilanes, and aminosilanes. A silane contains hydrogen and/or carbon groups, but does not contain a halogen. A polysilane may have the formula $\text{(H.sub.3Si—(SiH.sub.2).sub.n—SiH.sub.3)}$, where $n \geq 1$. Examples of silanes include silane (SiH.sub.4), disilane (Si.sub.2H.sub.6), trisilane, tetrasilane and organo silanes such as methylsilane, ethylsilane, isopropylsilane, t-butylsilane, dimethylsilane, diethylsilane, di-t-butylsilane, allylsilane, sec-butylsilane, hexylsilane, isoamylsilane, t-butylidisilane, di-t-butylidisilane, tetra-ethyl-ortho-silicate (also known as tetra-ethoxy-silane or TEOS) and the like.

(53) Example silicon-containing reactants include polysilanes ($\text{H.sub.3Si}-(\text{SiH.sub.2}).\text{sub.n}-\text{SiH.sub.3}$), where $n \geq 1$, such as silane, disilane, trisilane, tetrasilane; and trisilylamine:

(54) ##STR00001##

(55) In some embodiments, the silicon-containing reactant is an alkoxysilane. Alkoxysilanes that may be used include, but are not limited to, the following: $\text{H.sub.x}-\text{Si}-(\text{OR}).\text{sub.y}$ where $x=1-3$, $x+y=4$ and R is a substituted or unsubstituted alkyl group; and $\text{H.sub.x}(\text{RO}).\text{sub.y}-\text{Si}-\text{Si}-(\text{OR}).\text{sub.yH.sub.x}$ where $x=1-2$, $x+y=3$ and R is a substituted or unsubstituted alkyl group.

(56) Examples of silicon-containing reactants include: methylsilane; trimethylsilane (3MS); ethylsilane; butasilanes; pentasilanes; octasilanes; heptasilane; hexasilane; cyclobutasilane; cycloheptasilane; cyclohexasilane; cyclooctasilane; cyclopentasilane; 1,4-dioxa-2,3,5,6-tetrasilacyclohexane; diethoxymethylsilane (DEMS); diethoxysilane (DES); dimethoxymethylsilane; dimethoxysilane (DMOS); methyl-diethoxysilane (MDES); methyl-dimethoxysilane (MDMS); octamethoxydodecasiloxane (OMODDS); tert-butoxydisilane; tetramethylcyclotetrasiloxane (TMCTS); tetraoxymethylcyclotetrasiloxane (TOMCTS); triethoxysilane (TES); triethoxysiloxane (TRIES); and trimethoxysilane (TMS or TriMOS).

(57) In some embodiments, the silicon-containing precursor may be an aminosilane, with hydrogen atoms, such as bisdiethylaminosilane, diisopropylaminosilane, tert-butylamino silane (BTBAS), or tris(dimethylamino)silane (3DMAS). Aminosilane precursors include, but are not limited to, the following: $\text{H.sub.x}-\text{Si}-(\text{NR}).\text{sub.y}$ where $x=1-3$, $x+y=4$ and R is an organic or hydride group.

(58) In some embodiments, a halogen-containing silane or halosilane may be used such that the silane includes at least one hydrogen atom. A halosilane may have a formula SiX.sub.aH.sub.y whereby $X=\text{Cl, F, I, or Br}$, and $a+y=4$, where $a \geq 1$. A halosilane may have a formula $\text{SiX.sub.aH.sub.y}(\text{CH.sub.3}).\text{sub.z}$ where $X=\text{Cl, F, I, or Br}$, and $a+y+z=4$, where $a \geq 1$. Examples of halosilanes are iodosilanes, bromosilanes, chlorosilanes and fluorosilanes. Although halosilanes, particularly fluorosilanes, may form reactive halide species that can etch silicon materials, in certain embodiments described herein, the silicon-containing precursor is not present when a plasma is struck. Specific chlorosilanes are tetrachlorosilane (SiCl.sub.4), trichlorosilane (HSiC.sub.3), dichlorosilane (H.sub.2SiCl.sub.2), monochlorosilane (ClSiH.sub.3), chloroallylsilane, chloromethylsilane, dichloromethylsilane, chlorodimethylsilane, chloroethylsilane, t-butylchlorosilane, di-t-butylchlorosilane, chloroisopropylsilane, chloro-sec-butylsilane, t-butyl dimethylchlorosilane, thexyldimethylchlorosilane, monochlorotrimethylsilane, and the like.

(59) Examples of silicon-containing reactants include siloxanes, alkyl silane or hydrocarbon-substituted silane, or a nitrogen-containing carbon-containing reactant. Example siloxanes include linear siloxanes, cyclic siloxanes, and caged siloxanes. Example siloxanes include 2,4,6,8-tetramethylcyclotetrasiloxane (TMCTS), heptamethylcyclotetrasiloxane (HMCTS), silsesquioxane, disiloxanes, such as pentamethyldisiloxane (PMDSO) and tetramethyldisiloxane (TMDSO), and trisiloxanes such as hexamethyltrisiloxane, heptamethyltrisiloxane. Alkyl silanes include a central silicon atom with one or more alkyl groups bonded to it as well as one or more hydrogen atoms bonded to it. In some embodiments, any one or more of the alkyl groups contain 1-5 carbon atoms. The hydrocarbon groups may be saturated or unsaturated (e.g., alkene (e.g., vinyl), alkyne, and aromatic groups). Examples include but are not limited to trimethylsilane (3MS), triethylsilane, pentamethyl disilamethane ($(\text{CH.sub.3}).\text{sub.2Si}-\text{CH.sub.2}-\text{Si}(\text{CH.sub.3}).\text{sub.3}$), and dimethylsilane (2MS). Additionally, disilanes, trisilanes, or other higher silanes may be used in place of monosilanes. In some embodiments, one of the silicon atoms can have a carbon-containing or hydrocarbon group attached to it, and one of the silicon atoms can have a hydrogen atom attached to it. Example carbon-containing reactants including a nitrogen include methyl-substituted disilazanes and trisilazanes, such as tetramethyldisilazane and hexamethyl trisilazane.

(60) Examples of silicon-containing reactants include siloxanes such as cyclotetrasiloxanes such as heptamethylcyclotetrasiloxane (HMCTS) and tetramethylcyclotetrasiloxane. Other cyclic siloxanes

can also include but are not limited to cyclotrisiloxanes and cyclopentasiloxanes. For depositing oxygen doped silicon carbide films, other examples of suitable precursors include linear siloxanes such as, but not limited to, disiloxanes, such as pentamethyldisiloxane (PMDSO), tetramethyldisiloxane (TMDSO), hexamethyl trisiloxane, and heptamethyl trisiloxane. For undoped silicon carbide, examples of suitable precursors include monosilanes substituted with one or more alkyl, alkene, and/or alkyne groups containing, e.g., 1-5 carbon atoms. Examples include but are not limited to trimethylsilane (3MS), dimethylsilane (2MS), triethylsilane (TES), and pentamethyldisilamethane. Additionally, disilanes, trisilanes, or other higher silanes may be used in place of monosilanes. An example of one such disilane from the alkyl silane class is hexamethyldisilane (HMDS). Another example of a disilane from the alkyl silane class can include pentamethyldisilane (PMDS). Other types of alkyl silanes can include alkylcarbosilanes, which can have a branched polymeric structure with a carbon bonded to a silicon atom as well as alkyl groups bonded to a silicon atom. Examples include dimethyl trimethylsilyl methane (DTMSM) and bis-dimethylsilyl ethane (BDMSE). For depositing nitrogen doped silicon carbide (SiNC) films, examples of suitable precursors include, e.g., alkyldisilazanes and possibly compounds including amino (—NH.sub.2) and alkyl groups separately bonded to one or more silicon atoms. Alkyldisilazanes include silizanes and alkyl groups bonded to two silicon atoms. An example includes 1,1,3,3-tetramethyldisilazane (TMDSN). Inert gas may be provided along with the silicon-containing reactant. In one example, argon is provided at a rate of about 1-20 slm, and nitrogen (e.g., N.sub.2) is provided at a rate of about 0 slm to about 30 slm. In addition, hydrogen (H.sub.2) may be provided at a flow rate of about 0 slm to about 5 slm. The pressure in the reaction chamber may be maintained about 0.6 Torr to about 20 Torr. The substrate may be positioned on a temperature-controlled substrate support, which may be maintained at a temperature of about 150° C. to about 650° C.

(61) During the purge operation and the second purge operation, the reaction chamber may be evacuated and/or swept with non-reactive gas to remove non-adsorbed reactants and reaction byproducts. Other gases may be present in the purge gas, as well. In one example, the purge gas includes argon at a flow rate of about 1 slm to about 20 slm, nitrogen (e.g., N.sub.2) at a flow rate of about 0 slm to about 30 slm, hydrogen (e.g., H.sub.2) at a flow rate of about 0 slm to about 5 slm, oxygen (e.g., O.sub.2) at a flow rate of about 0 slm to about 5 slm, and nitrous oxide (e.g., N.sub.2O) at a flow rate of about 0 slm to about 5 slm. The chamber pressure and substrate support temperature may be as described above in relation to the dose operation.

(62) During the oxidation operation, the surface of the substrate is oxidized and cleaned with RF plasma. Gases may flow to the reaction chamber, including at least one oxygen-containing reactant. Example oxygen-containing reactants include, but are not limited to, oxygen (O.sub.2), nitrous oxide (N.sub.2O), ozone (O.sub.3), carbon monoxide (CO), carbon dioxide (CO.sub.2), etc. Other gases may be provided, as well. In one example, the following gases may be provided during the oxidation operation: argon at a flow rate of about 1-20 slm, nitrogen (e.g., N.sub.2) at a flow rate of about 0 slm to about 30 slm, hydrogen (e.g., H.sub.2) at a flow rate of about 0 slm to about 5 slm, oxygen (e.g., O.sub.2) at a flow rate of about 0.5 slm to about 5 slm, and nitrous oxide (e.g., N.sub.2O) at a flow rate of about 0 slm to about 5 slm. RF power may be provided in the range of about 500 W to about 6000 W for a 4-station chamber. The pressure and temperature may be as described above in relation to the dose operation.

(63) In cases where an inhibition operation is used (e.g., see inhibition operation **110** of FIG. **1C**), one or more of the following conditions may be used during the inhibition operation. An inhibition gas may be provided to the reaction chamber at a flow rate between about 10 sccm to about 300 sccm. Examples of inhibition gas species include nitrogen species, halogen species, and hydrogen species. However, any species that adsorb onto, react with, or otherwise interact with the substrate surface to passivate it and inhibit subsequent deposition may be used. In various embodiments, the inhibition operation may be a plasma-based inhibition or a thermally-based inhibition. During a

plasma-based inhibition, the structure is exposed to a plasma generated from an inhibition gas to form inhibitor species, also referred to as inhibition species. Examples of inhibition gases include nitrogen-containing gases, halogen-containing gases, and hydrogen-containing gases. Specific examples include nitrogen trifluoride (NF₃), molecular nitrogen (N₂), molecular hydrogen (H₂), ammonia (NH₃), amines, diols, diamines, aminoalcohols, thiols or combinations thereof. In some embodiments, the inhibition gas is a fluorine-containing gas, such as NF₃, CH_xF_y where x+y=4 and y is an integer greater of equal to 1, and sulfur hexafluoride (SF₆). In some embodiments, the inhibition operation may be a thermal-based inhibition process. For example, the inhibition operation may be performed by exposing the structure to a gas that includes an inhibition species under conditions where the inhibition species reacts with the surface of the substrate (e.g., without exposing the substrate to plasma during the inhibition operation).

(64) In some embodiments, an inhibition plasma is generated from a halogen-containing gas. Examples include NF₃, CH₃F, CH₂F₂, CHF₃, CF₄, SF₆, CH₃Cl, CH₂Cl₂, CHCl₃, and CCl₄. This results in —F, —Cl, or other halogen terminated surfaces throughout the depth of the gap (but most particularly near the top of the sidewalls) that passivate the surface and inhibit subsequent deposition. The halogen-containing gas or other inhibition gas may be about 0.5% to about 10%, about 1 to about 5%, or about 2% of the total volumetric flow to the chamber or other plasma generation space during the inhibition operation **110**, with the remaining flow being an inert gas such as N₂, Ar, and He.

(65) In cases where plasma is used, the plasma may be generated at an RF power level about 500 W to about 2000 W for a 4-station chamber. The substrate may be exposed to the plasma for a duration of about 0.1 second to about 10 seconds. The substrate support temperature may be about 200° C. to about 650° C. The pressure in the reaction chamber may be about 0.6 Torr to about 10 Torr.

(66) In cases where an etch operation is used (e.g., see etch operation **120** in FIG. **1D**), one or more of the following conditions may be used during the etch operation. An etch gas may be provided to the reaction chamber at a flow rate of about 50 sccm to about 500 sccm. In one example, the etch gas includes NF₃. In other cases, other etch gases may be used. In various embodiments, the etch gas may include nitrogen and/or a halogen such as fluorine or chlorine. Other gases may be provided to the reaction chamber as well, for example one or more of the gases described above in relation to the inhibition operation **110**. Generally, many of the same gases may be used for inhibition and etching, with lower flow rates typically resulting in inhibition and higher flow rates typically resulting in etching. The substrate may be exposed to plasma generated from the etch gas and/or other gases. The plasma may be generated at an RF power level about 750 W to about 8000 W for a 4-station chamber. The substrate may be exposed to the plasma for a duration of about 1 second to about 300 seconds. The substrate support temperature may be about 200° C. to about 650° C. The pressure in the reaction chamber may be about 0.6 Torr to about 2 Torr.

(67) In cases where deposition occurs through chemical vapor deposition (e.g., see FIG. **1E**), one or more of the following conditions may be used. The substrate may be simultaneously exposed to a first reactant (e.g., a silicon-containing reactant such as any of those described above) and to a second reactant (e.g., an oxygen-containing reactant such as any of those described above). At the same time, the substrate may be exposed to an energy source (e.g., thermal energy and/or plasma energy) to drive a gas phase reaction between the first and second reactants. This reaction produces a silicon-containing material, which is deposited on the substrate surface. The silicon-containing reactant may be provided at a flow rate about 150 sccm to about 900 sccm. The oxygen-containing reactant may be provided at a flow rate about 500 sccm to about 20,000 sccm. The pressure in the reaction chamber may be maintained about 2 Torr to about 20 Torr. Where the reaction is driven by thermal energy, the substrate support may be maintained at a temperature about 550° C. to about 650° C. Where the reaction is driven by plasma energy, the substrate support may be maintained at

a temperature of about 200° C. to about 650° C., and the plasma may be generated at an RF power level of about 750 W to about 5000 W for a 4-station chamber. The substrate may be exposed to the first and second reactants, and to the energy source driving the reaction, for a duration of about 0.1 second to about 10 seconds during each cycle (e.g., each cycle including a reactant delivery operation **130** and an impurity reduction operation **105**, as shown in FIG. **1E**).

Apparatus

(68) The embodiments herein may be practiced on any appropriate apparatus. An appropriate apparatus will include at least a processing chamber, a plasma generator for producing plasma in the processing chamber, and a controller that is configured to cause one or more of the methods described herein.

(69) FIG. **2** schematically shows an embodiment of a process station **200** that may be used to deposit material using atomic layer deposition (ALD), and/or chemical vapor deposition (CVD), either of which may be driven by thermal energy or plasma energy. The process station **200** may be used to practice any of the embodiments described herein, including e.g., those shown in FIGS. **1A-IE**. For simplicity, the process station **200** is depicted as a standalone process station having a process chamber body **202** for maintaining a low-pressure environment. However, it will be appreciated that a plurality of process stations **200** may be included in a common process tool environment. Further, it will be appreciated that, in some embodiments, one or more hardware parameters of process station **200**, including those discussed in detail below, may be adjusted programmatically by one or more computer controllers.

(70) Process station **200** fluidly communicates with reactant delivery system **201** for delivering process gases to a distribution showerhead **206**. Reactant delivery system **201** includes an optional mixing vessel **204** for blending and/or conditioning process gases for delivery to showerhead **206**. One or more mixing vessel inlet valves **220** may control introduction of process gases to mixing vessel **204**. Similarly, a showerhead inlet valve **205** may control introduction of process gasses to the showerhead **206**. In another example, the reactant delivery system **201** may maintain the reactants separate from one another until delivery to inside the chamber body **202**.

(71) Some reactants, like BTBAS, may be stored in liquid form prior to vaporization at and subsequent delivery to the process station. For example, the embodiment of FIG. **2** includes a vaporization point **203** for vaporizing liquid reactant to be supplied to mixing vessel **204**. In some embodiments, vaporization point **203** may be a heated vaporizer. The reactant vapor produced from such vaporizers may condense in downstream delivery piping. Exposure of incompatible gases to the condensed reactant may create small particles. These small particles may clog piping, impede valve operation, contaminate substrates, etc. Some approaches to addressing these issues involve sweeping and/or evacuating the delivery piping to remove residual reactant. However, sweeping the delivery piping may increase process station cycle time, degrading process station throughput. Thus, in some embodiments, delivery piping downstream of vaporization point **203** may be heat traced. In some examples, mixing vessel **204** may also be heat traced. In one non-limiting example, piping downstream of vaporization point **203** has an increasing temperature profile extending from approximately 100° C. to approximately 150° C. at mixing vessel **204**.

(72) In some embodiments, reactant liquid may be vaporized at a liquid injector. For example, a liquid injector may inject pulses of a liquid reactant into a carrier gas stream upstream of the mixing vessel. In one scenario, a liquid injector may vaporize reactant by flashing the liquid from a higher pressure to a lower pressure. In another scenario, a liquid injector may atomize the liquid into dispersed microdroplets that are subsequently vaporized in a heated delivery pipe. It will be appreciated that smaller droplets may vaporize faster than larger droplets, reducing a delay between liquid injection and complete vaporization. Faster vaporization may reduce a length of piping downstream from vaporization point **203**. In one scenario, a liquid injector may be mounted directly to mixing vessel **204**. In another scenario, a liquid injector may be mounted directly to showerhead **206**.

(73) In some embodiments, a liquid flow controller upstream of vaporization point **203** may be provided for controlling a mass flow of liquid for vaporization and delivery to process station **200**. For example, the liquid flow controller (LFC) may include a thermal mass flow meter (MFM) located downstream of the LFC. A plunger valve of the LFC may then be adjusted responsive to feedback control signals provided by a proportional-integral-derivative (PID) controller in electrical communication with the MFM. However, it may take one second or more to stabilize liquid flow using feedback control. This may extend a time for dosing a liquid reactant. Thus, in some embodiments, the LFC may be dynamically switched between a feedback control mode and a direct control mode. In some embodiments, the LFC may be dynamically switched from a feedback control mode to a direct control mode by disabling a sense tube of the LFC and the PID controller.

(74) Showerhead **206** distributes process gases toward substrate **212**. In the embodiment shown in FIG. 2, substrate **212** is located beneath showerhead **206**, and is shown resting on a pedestal **208**. It will be appreciated that showerhead **206** may have any suitable shape, and may have any suitable number and arrangement of ports for distributing processes gases to substrate **212**.

(75) In some embodiments, a microvolume **207** is located beneath showerhead **206**. Performing an ALD and/or CVD process in a microvolume rather than in the entire volume of a process station may reduce reactant exposure and sweep times, may reduce times for altering process conditions (e.g., pressure, temperature, etc.), may limit an exposure of process station robotics to process gases, etc. Example microvolume sizes include, but are not limited to, volumes between 0.1 liter and 2 liters. This microvolume also impacts productivity throughput. While deposition rate per cycle drops, the cycle time also simultaneously reduces. In certain cases, the effect of the latter is dramatic enough to improve overall throughput of the module for a given target thickness of film.

(76) In some embodiments, pedestal **208** may be raised or lowered to expose substrate **212** to microvolume **207** and/or to vary a volume of microvolume **207**. For example, in a substrate transfer phase, pedestal **208** may be lowered to allow substrate **212** to be loaded onto pedestal **208**. During a deposition process phase, pedestal **208** may be raised to position substrate **212** within microvolume **207**. In some embodiments, microvolume **207** may completely enclose substrate **212** as well as a portion of pedestal **208** to create a region of high flow impedance during a deposition process.

(77) Optionally, pedestal **208** may be lowered and/or raised during portions the deposition process to modulate process pressure, reactant concentration, etc., within microvolume **207**. In one scenario where process chamber body **202** remains at a base pressure during the deposition process, lowering pedestal **208** may allow microvolume **207** to be evacuated. Example ratios of microvolume to process chamber volume include, but are not limited to, volume ratios about 1:200 to about 1:10. It will be appreciated that, in some embodiments, pedestal height may be adjusted programmatically by a suitable computer controller.

(78) In another scenario, adjusting a height of pedestal **208** may allow a plasma density to be varied during plasma activation and/or treatment cycles included in the deposition process. At the conclusion of the deposition process phase, pedestal **208** may be lowered during another substrate transfer phase to allow removal of substrate **212** from pedestal **208**.

(79) While the example microvolume variations described herein refer to a height-adjustable pedestal, it will be appreciated that, in some embodiments, a position of showerhead **206** may be adjusted relative to pedestal **208** to vary a volume of microvolume **207**. Further, it will be appreciated that a vertical position of pedestal **208** and/or showerhead **206** may be varied by any suitable mechanism within the scope of the present disclosure. In some embodiments, pedestal **208** may include a rotational axis for rotating an orientation of substrate **212**. It will be appreciated that, in some embodiments, one or more of these example adjustments may be performed programmatically by one or more suitable computer controllers.

(80) Returning to the embodiment shown in FIG. 2, showerhead **206** and pedestal **208** electrically communicate with RF power supply **214** and matching network **216** for powering a plasma. In

some embodiments, the plasma energy may be controlled by controlling one or more of a process station pressure, a gas concentration, an RF source power, an RF source frequency, and a plasma power pulse timing. For example, RF power supply **214** and matching network **216** may be operated at any suitable power to form a plasma having a desired composition of radical species. Examples of suitable powers are included above. Likewise, RF power supply **214** may provide RF power of any suitable frequency. In some embodiments, RF power supply **214** may be configured to control high- and low-frequency RF power sources independently of one another. Example low-frequency RF frequencies may include, but are not limited to, frequencies about 50 kHz to about 200 kHz. Example high-frequency RF frequencies may include, but are not limited to, frequencies about 1.8 MHz to about 2.45 GHz. It will be appreciated that any suitable parameters may be modulated discretely or continuously to provide plasma energy for the surface reactions. In one non-limiting example, the plasma power may be intermittently pulsed to reduce ion bombardment with the substrate surface relative to continuously powered plasmas.

(81) In some embodiments, the plasma may be monitored in-situ by one or more plasma monitors. In one scenario, plasma power may be monitored by one or more voltage, current sensors (e.g., VI probes). In another scenario, plasma density and/or process gas concentration may be measured by one or more optical emission spectroscopy sensors (OES). In some embodiments, one or more plasma parameters may be programmatically adjusted based on measurements from such in-situ plasma monitors. For example, an OES sensor may be used in a feedback loop for providing programmatic control of plasma power. It will be appreciated that, in some embodiments, other monitors may be used to monitor the plasma and other process characteristics. Such monitors may include, but are not limited to, infrared (IR) monitors, acoustic monitors, and pressure transducers.

(82) In some embodiments, the plasma may be controlled via input/output control (IOC) sequencing instructions. In one example, the instructions for setting plasma conditions for a plasma process phase may be included in a corresponding plasma activation recipe phase of a deposition process recipe or an impurity reduction process recipe. In some cases, process recipe phases may be sequentially arranged, so that all instructions for a deposition process phase are executed concurrently with that process phase. In some embodiments, instructions for setting one or more plasma parameters may be included in a recipe phase preceding a plasma process phase. For example, a first recipe phase may include instructions for setting a flow rate of an inert and/or a reactant gas, instructions for setting a plasma generator to a power set point, and time delay instructions for the first recipe phase. A second, subsequent recipe phase may include instructions for enabling the plasma generator and time delay instructions for the second recipe phase. A third recipe phase may include instructions for disabling the plasma generator and time delay instructions for the third recipe phase. It will be appreciated that these recipe phases may be further subdivided and/or iterated in any suitable way within the scope of the present disclosure.

(83) In some deposition processes, plasma strikes last on the order of a few seconds or more in duration. In certain implementations, much shorter plasma strikes may be used. These may be on the order of 10 ms to 1 second, typically, about 20 to 80 ms, with 50 ms being a specific example. Such very short RF plasma strikes require extremely quick stabilization of the plasma. To accomplish this, the plasma generator may be configured such that the impedance match is set preset to a particular voltage, while the frequency is allowed to float. High-frequency plasmas may be generated at an RF frequency at about 13.56 MHz. In various embodiments disclosed herein, the frequency is allowed to float to a value that is different from this standard value. By permitting the frequency to float while fixing the impedance match to a predetermined voltage, the plasma can stabilize much more quickly, a result which may be important when using the very short plasma strikes associated with some types of deposition cycles.

(84) In some embodiments, pedestal **208** may be temperature controlled via heater **210**. Further, in some embodiments, pressure control for deposition process station **200** may be provided by butterfly valve **218**. As shown in the embodiment of FIG. 2, butterfly valve **218** throttles a vacuum

provided by a downstream vacuum pump (not shown). However, in some embodiments, pressure control of process station **200** may also be adjusted by varying a flow rate of one or more gases introduced to process station **200**.

(85) One or more process stations may be included in a multi-station processing tool. FIG. **3** shows a schematic view of an embodiment of a multi-station processing tool **300** with an inbound load lock **302** and an outbound load lock **304**, either or both of which may include a remote plasma source. A robot **306**, at atmospheric pressure, is configured to move substrates or wafers from a cassette loaded through a pod **308** into inbound load lock **302** via an atmospheric port **310**. A substrate is placed by the robot **306** on a pedestal **312** in the inbound load lock **302**, the atmospheric port **310** is closed, and the load lock is pumped down. Where the inbound load lock **302** includes a remote plasma source, the substrate may be exposed to a remote plasma treatment in the load lock prior to being introduced into a processing chamber **314**. Further, the substrate also may be heated in the inbound load lock **302** as well, for example, to remove moisture and adsorbed gases. Next, a chamber transport port **316** to processing chamber **314** is opened, and another robot (not shown) places the substrate into the reactor on a pedestal of a first station shown in the reactor for processing. While the embodiment depicted in FIG. **3** includes load locks, it will be appreciated that, in some embodiments, direct entry of a substrate into a process station may be provided. In various embodiments, the soak gas is introduced to the station when the substrate is placed by the robot **306** on the pedestal **312**.

(86) The depicted processing chamber **314** includes four process stations, numbered from **1** to **4** in the embodiment shown in FIG. **3**. Each station has a heated pedestal (shown at **318** for station **1**), and gas line inlets. It will be appreciated that in some embodiments, each process station may have different or multiple purposes. For example, in some embodiments, a process station may be switchable between an ALD and PEALD process mode. Additionally or alternatively, in some embodiments, processing chamber **314** may include one or more matched pairs of ALD and plasma-enhanced ALD process stations. While the depicted processing chamber **314** includes four stations, it will be understood that a processing chamber according to the present disclosure may have any suitable number of stations. For example, in some embodiments, a processing chamber may have five or more stations, while in other embodiments a processing chamber may have three or fewer stations.

(87) FIG. **3** depicts an embodiment of a wafer handling system **390** for transferring substrates within processing chamber **314**. In some embodiments, wafer handling system **390** may transfer substrates between various process stations and/or between a process station and a load lock. It will be appreciated that any suitable wafer handling system may be employed. Non-limiting examples include wafer carousels and wafer handling robots. FIG. **3** also depicts an embodiment of a system controller **350** employed to control process conditions and hardware states of process tool **300**. System controller **350** may include one or more memory devices **356**, one or more mass storage devices **354**, and one or more processors **352**. Processor **352** may include a CPU or computer, analog and/or digital input/output connections, stepper motor controller boards, etc. In some embodiments, system controller **350** includes machine-readable instructions for performing operations such as those described herein.

(88) In some embodiments, system controller **350** controls the activities of process tool **300**. System controller **350** executes system control software **358** stored in mass storage device **354**, loaded into memory device **356**, and executed on processor **352**. Alternatively, the control logic may be hard coded in the system controller **350**. Applications Specific Integrated Circuits, Programmable Logic Devices (e.g., field-programmable gate arrays, or FPGAs) and the like may be used for these purposes. In the following discussion, wherever “software” or “code” is used, functionally comparable hard coded logic may be used in its place. System control software **358** may include instructions for controlling the timing, mixture of gases, amount of gas flow, chamber and/or station pressure, chamber and/or station temperature, substrate temperature, target power

levels, RF power levels, substrate pedestal, chuck and/or susceptor position, and other parameters of a particular process performed by process tool **300**. System control software **358** may be configured in any suitable way. For example, various process tool component subroutines or control objects may be written to control operation of the process tool components used to carry out various process tool processes. System control software **358** may be coded in any suitable computer readable programming language.

(89) Broadly speaking, the controller may be defined as electronics having various integrated circuits, logic, memory, and/or software that receive instructions, issue instructions, control operation, enable cleaning operations, enable endpoint measurements, and the like. The integrated circuits may include chips in the form of firmware that store program instructions, digital signal processors (DSPs), chips defined as application specific integrated circuits (ASICs), and/or one or more microprocessors, or microcontrollers that execute program instructions (e.g., software). Program instructions may be instructions communicated to the controller in the form of various individual settings (or program files), defining operational parameters for carrying out a particular process on or for a semiconductor wafer or to a system. The operational parameters may, in some embodiments, be part of a recipe defined by process engineers to accomplish one or more processing steps during the fabrication of one or more layers, materials, metals, oxides, silicon, silicon dioxide, surfaces, circuits, and/or dies of a wafer.

(90) The controller, in some implementations, may be a part of or coupled to a computer that is integrated with, coupled to the system, otherwise networked to the system, or a combination thereof. For example, the controller may be in the “cloud” or all or a part of a fab host computer system, which can allow for remote access of the wafer processing. The computer may enable remote access to the system to monitor current progress of fabrication operations, examine a history of past fabrication operations, examine trends or performance metrics from a plurality of fabrication operations, to change parameters of current processing, to set processing steps to follow a current processing, or to start a new process. In some examples, a remote computer (e.g. a server) can provide process recipes to a system over a network, which may include a local network or the Internet. The remote computer may include a user interface that enables entry or programming of parameters and/or settings, which are then communicated to the system from the remote computer. In some examples, the controller receives instructions in the form of data, which specify parameters for each of the processing steps to be performed during one or more operations. It should be understood that the parameters may be specific to the type of process to be performed and the type of tool that the controller is configured to interface with or control. Thus as described above, the controller may be distributed, such as by including one or more discrete controllers that are networked together and working towards a common purpose, such as the processes and controls described herein. An example of a distributed controller for such purposes would be one or more integrated circuits on a chamber in communication with one or more integrated circuits located remotely (such as at the platform level or as part of a remote computer) that combine to control a process on the chamber.

(91) Without limitation, example systems may include a plasma etch chamber or module, a deposition chamber or module, a spin-rinse chamber or module, a metal plating chamber or module, a clean chamber or module, a bevel edge etch chamber or module, a physical vapor deposition (PVD) chamber or module, a chemical vapor deposition (CVD) chamber or module, an atomic layer deposition (ALD) chamber or module, an atomic layer etch (ALE) chamber or module, an ion implantation chamber or module, a track chamber or module, and any other semiconductor processing systems that may be associated or used in the fabrication and/or manufacturing of semiconductor wafers.

(92) As noted above, depending on the process step or steps to be performed by the tool, the controller might communicate with one or more of other tool circuits or modules, other tool components, cluster tools, other tool interfaces, adjacent tools, neighboring tools, tools located

throughout a factory, a main computer, another controller, or tools used in material transport that bring containers of wafers to and from tool locations and/or load ports in a semiconductor manufacturing factory.

Experimental

(93) Over the course of several experiments, the inventors have confirmed that the methods described herein may be used to substantially reduce impurities (e.g., F, C, H, and/or N) in silicon oxide film. These results show significant improvement (e.g., one to two orders of magnitude greater reduction in fluorine concentration) compared to previous standard passivation methods, even at much shorter processing times.

(94) FIGS. 4A-4C show the concentration of fluorine at different depths for silicon oxide films deposited according to some methods. FIG. 4A shows results related to a silicon oxide film deposited using ALD only (e.g., similar to the methods shown in FIGS. 1A and 1B, but without the impurity reduction operation 105). In the ALD-only example of FIG. 4A, the fluorine concentration in the silicon oxide film is on the order of about $1E17$ atoms/cc. The results in FIG. 4A show not only the fluorine concentration in the film, but the concentration of carbon, hydrogen, nitrogen, silicon, and oxygen, as well. FIG. 4B shows results related to three silicon oxide films deposited using a deposition-etch-deposition scheme that also utilized a periodic passivation operation without impurity reduction (e.g., similar to the method shown in FIG. 1D, but using a passivation operation without impurity reduction instead of impurity reduction operation 105), as well as one film deposited without any etch or passivation operation. In the example of FIG. 4B, the fluorine concentration in the silicon oxide films is on the order of about $1E18$ - $1E19$ atoms/cc, and is highest at the interfaces/depths where the deposition was stopped, e.g., to perform the etch operation. FIG. 4C shows results related to two silicon oxide films 401 and 402 deposited using a deposition-etch-deposition scheme that also utilized a periodic passivation operation without impurity reduction, where the deposition operations were driven by high density plasma (HDP). In this example, the fluorine concentration in the films were on the order of about $5E19$ atoms/cc.

(95) FIGS. 5A and 5B show the concentration of fluorine, carbon, hydrogen, nitrogen, silicon, and oxygen at different depths in silicon oxide films deposited according to various techniques described herein. In FIG. 5A, the silicon oxide film was deposited using continuous thermal CVD, without any impurity reduction operation. In FIG. 5B, the silicon oxide film was deposited using the modified CVD technique described in relation to FIG. 1E, with an impurity reduction operation performed periodically throughout the deposition. The impurity reduction operation was alternated with the chemical vapor deposition operation. The results in FIGS. 5A and 5B are summarized in FIG. 6 and in Tables 1 and 2, below.

(96) FIG. 6 shows the concentration of carbon, hydrogen, and nitrogen in silicon oxide films deposited as described in relation to FIGS. 5A and 5B. Process A relates to the results shown in FIG. 5A, where the deposition occurred according to a continuous thermal chemical vapor deposition technique that did not involve an impurity reduction operation. Process B relates to the results shown in FIG. 5B, where the deposition occurred according to the modified chemical vapor deposition technique that involved an impurity reduction operation performed periodically during the deposition, as described in relation to FIG. 1E. The reported concentrations in FIG. 6 are the average concentrations, excluding the very top and very bottom edges of the silicon oxide film. As shown in FIG. 6, the impurity reduction operation had a substantial effect on the hydrogen concentration in the silicon oxide film (e.g., reducing from about 7.70% H to about 0.52% H), as well as the carbon concentration in the film (e.g., reducing from about 2.68% C to about 0.12% C). The impurity reduction operation also had a noticeable effect on the nitrogen concentration in the silicon oxide film (e.g., reducing from about 1.88% N to about 0.97% N), though this effect wasn't as substantial as the effect on the hydrogen and carbon concentrations. The values reported in FIG. 6 were based on RBS and SIMS data. Process A produced a silicon oxide film that was about 30.3% silicon and about 62.0% oxygen (e.g., for a total of about 92.3% silicon and oxygen). By

contrast, process B produced a silicon oxide film that was about 33.3% silicon and 66.1% oxygen (e.g., for a total of about 99.4% silicon and oxygen). In other words, process B produced a substantially more pure film than process A.

(97) Table 1 shows the concentration of different impurities when comparing process A and process B from FIG. 6. To reiterate, process A relates to the results shown in FIG. 5A, and process B relates to the results shown in FIG. 5B. These results generally match those in FIG. 6, showing substantial reductions in hydrogen concentration and carbon concentration, as well as a more modest reduction in nitrogen concentration, in cases where an impurity reduction operation is used during deposition.

(98) TABLE-US-00001

Process A Impurity Concentration (atoms/cc)	Process B Impurity Concentration (atoms/cc)
H 1.53E21	1.13E20
C 5.33E20	2.66E19
N 3.74E20	2.12E20
F 4.96E+18	1.40E+16

(99) Table 2 presents the concentration of the different impurities shown in Table 1, reported in atomic % rather than atoms/cc. The values in Table 2 match those reported in FIG. 6, and include further information regarding the fluorine concentration in each film.

(100) TABLE-US-00002

Process A Impurity Concentration (atomic %)	Process B Impurity Concentration (atomic %)
H 7.70	0.52
C 2.68	0.12
N 1.88	0.97
F 0.3	0.00

(101) These results provided in this section can be used to quantitatively compare silicon oxide films deposited according to methods without impurity reduction and silicon oxide films deposited according to the disclosed embodiments. Generally speaking, the impurity reduction operation resulted in a 93% decrease in the concentration of hydrogen; a 95% decrease in the concentration of carbon; a 43% decrease in the concentration of nitrogen; and a 99.7% decrease in the concentration of fluorine. These impurity reductions were substantial.

(102) In various embodiments, the silicon oxide film may have any one or more of the following properties. The concentration of fluorine may be on the order of about 1E16 atoms/cc. In some cases, the fluorine concentration may be about 9E16 atoms/cc or less, or about 5E16 atoms/cc or less, or about 2E16 atoms/cc or less, or about 1E16 atoms/cc or less. The concentration of fluorine may be about 0.3% (atomic %) or less, or about ~1E-5% (atomic %) or less. The concentration of carbon may be about 9E19 atoms/cc or less, or about 5E19 atoms/cc or less. The concentration of carbon may be about 2% (atomic %) or less, or about 1% (atomic %), or about 0.5% (atomic %) or less, or about 0.25% (atomic %) or less. The concentration of nitrogen may be about 3E20 atoms/cc or less. The concentration of nitrogen may be about 1.5% (atomic %) or less, or about 1% (atomic %) or less.

(103) FIGS. 7A and 7B presents FTIR data confirming the periodic impurity reduction operation resulted in a substantial decrease in the concentration of hydrogen in the silicon oxide film (FIG. 7A), as well as an increase in silicon and oxygen concentrations in the silicon oxide film (FIG. 7B). In FIGS. 7A and 7B, “Std dep” refers to a deposition that did not involve an impurity reduction operation, while “periodic treatment” refers to a deposition that involved periodically performing an impurity reduction operation as described above. Hydrogen-related FTIR peaks would be expected at wavenumbers of about 3675 cm.sup.-1 (related to —OH stretching), 3300 cm.sup.-1 (related to —OH stretching), and 1615 cm.sup.-1 (related to H—O—H stretching). As shown in FIG. 7A, the deposition that did not utilize an impurity reduction operation showed noticeable peaks at each of these wavenumbers, indicating the presence of a significant amount of hydrogen in the silicon oxide film. By contrast, where the impurity reduction operation was used, these peaks essentially disappear, indicating that the concentration of hydrogen is much lower. Silicon- and oxygen-related peaks would be expected at wavenumbers of about 1200 cm.sup.-1 (related to Si—O stretching), 1075 cm.sup.-1 (related to Si—O—Si stretching), 815 cm.sup.-1 (related to Si—O bending), and 460 cm.sup.-1 (related to Si—O being out of plane). As shown in FIG. 7B, both the deposition without impurity reduction and the deposition involving the impurity reduction operation show peaks at these wavenumbers. However, the peaks are higher in the case where the impurity reduction operation was used, indicating that there is relatively more silicon and oxygen

present in the silicon oxide film. In other words, there are fewer impurities present, and the silicon oxide film is purer.

CONCLUSION

(104) Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, it will be apparent that certain changes and modifications may be practiced within the scope of the appended claims. It should be noted that there are many alternative ways of implementing the processes, systems, and apparatus of the present embodiments. Accordingly, the present embodiments are to be considered as illustrative and not restrictive, and the embodiments are not to be limited to the details given herein.

Claims

1. A method for depositing a doped or undoped silicon-containing film on a substrate, the method comprising: (a) exposing the substrate to a first reactant, wherein the first reactant is a silicon-containing reactant; (b) reacting at least the first reactant in a first plasma to form a doped or undoped silicon-containing material, and depositing a portion of the doped or undoped silicon-containing film on the substrate, the doped or undoped silicon-containing film having a first impurity concentration; (c) before the depositing of the portion of the doped or undoped silicon-containing film is complete, performing an impurity reduction operation comprising: (i) generating a second plasma from a plasma generation gas, wherein the plasma generation gas comprises inert gas and hydrogen (H.sub.2), and wherein the plasma generation gas is substantially free of oxygen (O.sub.2), and (ii) exposing the substrate to the second plasma to thereby reduce the first impurity concentration to a second impurity concentration in the doped or undoped silicon-containing material; and (d) repeating (a) with at least one of (b) or (c) until the doped or undoped silicon-containing film is deposited to a final thickness.
2. The method of claim 1, wherein the impurity is selected from the group consisting of fluorine, carbon, hydrogen, nitrogen, and combinations thereof.
3. The method of claim 1, further comprising (e) exposing the substrate to a second reactant, wherein reacting at least the first reactant comprises reacting the first reactant with the second reactant to form the doped or undoped silicon-containing material, wherein repeating (a) with at least one of (b) or (c) further comprises repeating (e).
4. The method of claim 3, wherein the first reactant and second reactant are introduced to a chamber comprising the substrate simultaneously.
5. The method of claim 3, wherein the second reactant is an oxygen-containing reactant.
6. The method of claim 3, wherein the first reactant and second reactant are introduced to a chamber comprising the substrate in temporally separated pulses.
7. The method of claim 1, wherein the doped or undoped silicon-containing film is deposited in recessed features formed in a surface of the substrate, the method further comprising exposing the substrate to etching chemistry to etch a top portion of the doped or undoped silicon-containing film in the recessed features before the doped or undoped silicon-containing film fully fills the recessed features.
8. The method of claim 1, wherein the doped or undoped silicon-containing film is deposited in recessed features formed in a surface of the substrate, the method further comprising exposing the substrate to inhibition chemistry to selectively inhibit deposition near a top of the recessed features as compared to a bottom and middle of the recessed features.
9. The method of claim 1, wherein the second impurity concentration is less than 10 times the first impurity concentration.
10. The method of claim 1, wherein concentration of fluorine in the doped or undoped silicon-containing film is on the order of about 1×10^{16} atoms/cc once the doped or undoped silicon-containing film reaches the final thickness.

11. The method of claim 1, wherein concentration of carbon in the doped or undoped silicon-containing film is about $1E20$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness.
 12. The method of claim 1, wherein concentration of carbon in the doped or undoped silicon-containing film is about 2% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness.
 13. The method of claim 1, wherein concentration of hydrogen in the doped or undoped silicon-containing film is about $5E20$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness.
 14. The method of claim 1, wherein concentration of hydrogen in the doped or undoped silicon-containing film is about 5% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness.
 15. The method of claim 1, wherein concentration of nitrogen in the doped or undoped silicon-containing film is about $3E20$ atoms/cc or less once the doped or undoped silicon-containing film reaches the final thickness.
 16. The method of claim 1, wherein concentration of nitrogen in the doped or undoped silicon-containing film is about 1.5% or less (atomic percent) once the doped or undoped silicon-containing film reaches the final thickness.
 17. The method of claim 1, wherein generating the plasma in (d)(i) comprises flowing the inert gas at a rate of about 2 slm to about 60 slm, flowing the hydrogen at a rate of about 0.5 slm to about 5 slm, and generating the plasma at an RF power level that includes about 1000 W to about 6000 W HF RF, and about 0 W to about 5000 W LF RF.
 18. The method of claim 1, wherein the plasma generation gas comprises nitrogen (N.sub.2).
 19. The method of claim 1, wherein the first reactant comprises a silane.
 20. An apparatus for depositing a doped or undoped silicon-containing film on a substrate, the apparatus comprising: a processing chamber; an inlet to the processing chamber for introducing reactants to the processing chamber; a plasma generator for generating a plasma in the processing chamber; and a controller configured to: cause the substrate to be exposed to a first reactant, wherein the first reactant is a silicon-containing reactant; cause reacting of at least the first reactant in a first plasma to form a doped or undoped silicon-containing material, and depositing a portion of the doped or undoped silicon-containing film on the substrate, the doped or undoped silicon-containing film having a first impurity concentration; before the depositing of the portion of the doped or undoped silicon-containing film is complete, cause an impurity reduction operation comprising: (i) causing generation of a second plasma from a plasma generation gas, wherein the plasma generation gas comprises inert gas and hydrogen (H.sub.2), and wherein the plasma generation gas is substantially free of oxygen (O.sub.2), and (ii) causing exposure of the substrate to the second plasma to thereby reduce the first impurity concentration to a second impurity concentration in the doped or undoped silicon-containing material; and (d) causing repeating of (a) with at least one of (b) or (c) until the doped or undoped silicon-containing film is deposited to a final thickness.
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